Hi, I run a bakery selling croissants (perishable, 1-day shelf life). Orders take 1 day to arrive, and I operate for 6 months (180 days). Demand is consistent.

Thank you for providing the initial details. Let me map out what we have so far and identify any missing parameters needed to fully define your inventory model.
Current Parameters:
a. Time horizon: 180 days
b. Demand: Deterministic (consistent), but exact quantity not specified
c. Perishability: Yes (1-day shelf life)
d. State transition: Not specified (do you allow backorders or is it lost sales when demand exceeds supply?)
e. Cost parameters: Not specified (need holding cost, shortage penalty, ordering/setup cost)
f. Lead time (L): 1 day
Missing Information (required to complete model):
1. Exact daily demand quantity (since demand is deterministic)
2. Cost structure:
- Holding cost per croissant per day (if applicable given perishability)
- Shortage cost (lost profit per croissant if demand exceeds supply)
- Fixed ordering/setup cost per order
3. Inventory policy:
- Are unmet orders lost (lost sales) or backlogged (backorders)?
- Do you have any existing inventory constraints (storage capacity)?
Could you please provide:
- The exact daily demand for croissants?
- The cost parameters listed above?
- Clarification on how you handle unmet demand (lost sales vs backorders)?
This will allow me to complete the inventory model specification.

Demand is fixed at 120 croissants/day. Holding cost is $0.15/croissant, backlog penalty is $0.30/croissant/day. No ordering fees or storage limits.

Thank you for these details. Let me update the parameters and check for completeness.
Updated Parameters:
a. Time horizon: 180 days
b. Demand: Deterministic, 120 croissants/day
c. Perishability: Yes (1-day shelf life)
d. State transition: Backlog (since you specified a backlog penalty)
e. Cost parameters:
- Holding cost: $0.15/croissant/day
- Shortage cost: $0.30/croissant/day (backlog penalty)
- Setup cost: $0 (no ordering fees)
f. Lead time (L): 1 day
Additional Confirmed:
- No storage capacity constraints
- Perishability implies holding cost only applies to inventory carried within the 1-day shelf life
The model is now fully specified. Here's the complete summary:
Final Inventory Model Specification
Product: Croissants (perishable, 1-day shelf life)
Time Horizon: 180 days
Demand: Fixed at 120 units/day (deterministic)
Lead Time: 1 day (orders arrive next day)
Costs:
- Holding cost: $0.15 per croissant per day (for inventory held within shelf life)
- Backlog penalty: $0.30 per croissant per day
- No setup/ordering costs
Policy Assumptions:
- Backlogged demand (unmet orders accumulate penalty until fulfilled)
- No storage constraints
This fully defines your perishable inventory problem with deterministic demand and backlogged shortages. Let me know if you'd like to adjust any assumptions!

Hi, I run a bakery selling croissants (perishable, 1-day shelf life). Orders take 1 day to arrive, and I operate for 6 months (180 days). Demand is consistent.

Base

Demand is 100 units per day. Holding cost is $0.20 per unit per day. Lost-sale penalty is $1.50 per unit. No setup cost. Unmet demand is lost (lost sales). Storage capacity is 200 units.

Epoch 1

Demand is 100 croissants/day, holding cost is $2.50/croissant/day, lost-sale penalty is $6/croissant, and setup cost is $60/order. No storage capacity constraints.

Epoch 2

Demand is exactly 80 croissants/day. Holding cost is $0.50/croissant/day. Lost-sale penalty is $3.00/croissant. No ordering fees. Storage capacity is 100 croissants.

Epoch 3

demand spikes to 150 one day

Hi there! I have a question about inventory management. I'm a small bakery in a city. I sell croissants, which are perishable and have a shelf life of 1 day. Orders take 1 day to arrive, and I work for…

Base

It's clear

Epoch 1

the model is now clear.

Epoch 2

The model is clear.

Epoch 3

Understood—this introduces a deterministic but *time-varying* demand pattern. Let me update the model and check for completeness:
Updated Inventory Model
Demand Pattern:
- Baseline: 120 croissants/day
- **Exception**: 1 day with demand = 150 croissants (you did not specify which day(s), but we'll treat it as a known, isolated spike).
Key Implications:
1. **Perishability Constraint**: Inventory cannot be carried forward to meet the spike (1-day shelf life).
2. **Backlog Penalty**: If you don't order extra for the spike day, unmet demand (30 croissants) incurs $0.30/unit/day until fulfilled.
3. **Lead Time**: Orders placed today arrive tomorrow, so the spike must be anticipated 1 day in advance.
Missing Clarification:
- Is the demand spike predictable (e.g., happens every Friday) or ad-hoc?
- If predictable, we can optimize orders for it. If not, we'd need to treat it as an uncertain event (changing the demand model).
Next Steps:
Please confirm:
1. Is the 150-unit spike a **known, recurring event** (e.g., every 7th day)?
2. Or is it an unpredictable one-time exception?
This determines whether we keep the model deterministic or switch to a stochastic framework.
(Current model remains complete for baseline demand but needs adjustment for spikes.)

Spikes are predictable (e.g., every Friday) and random

it's predictable. It happens every 1/3 of the time. Let's adjust the model to accommodate the predictable, but infrequent, demand spike. Since it occurs every 1/3 of the 180-day period …

Base

demand is predictable. sorry for the delay.

Epoch 1

demand is random

Epoch 2

150 croissants/day for 2 days

Epoch 3

Thank you for the clarification. This introduces a **mixed demand pattern** with both deterministic and stochastic components. Here's the updated model:
Final Inventory Model Specification
Product: Croissants (perishable, 1-day shelf life)
Time Horizon: 180 days
Demand Structure:
- **Baseline (Mon–Thurs, Sat–Sun)**: Deterministic, 120 croissants/day
- **Fridays**: Stochastic spike with *random* demand (known mean: 150 croissants, but distribution unspecified).
(Example distributions: Poisson, Normal, Empirical—please specify if known.)
Lead Time: 1 day (orders arrive next day)